# The Sizing And Trajectory Optimisation Of The Guest Transport Vehicle


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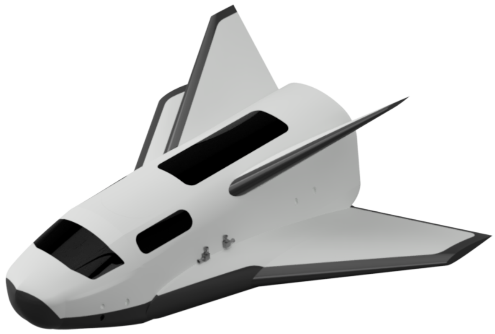
ISH06: Guest Transport Vehicle Design

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**Abstract**

This report details the sizing and optimisation of the re-entry trajectory for *Portal* - a guest transport vehicle designed for the *Imperial Skyview Space Hotel.* The driving design goal was to reduce the number of guest transport vehicles required over the lifespan of *Skyview.* This was achieved with an insulating thermal protection system, which imposed constraints on the maximum heat flux *Portal* can safely withstand. A 2-dimensional trajectory model alongside empirical methods for estimating heat fluxes was implemented for analysis of a given vehicle geometry to ensure these constraints were met. The fuselage was sized to house a luxurious passenger cabin, 5 m$^3$ of cargo, and other sub-systems. The main wing was sized to minimise the heat load on *Portal* whilst ensuring specified glided landing requirements. The stabilisers were sized to provide sufficient stability during the subsonic phase of flight. A genetic algorithm was used to find an angle of attack profile for the final geometry which minimised the error to a desired trajectory path in an attempt to minimise the heat load on *Portal* whilst adhering to applied constraints.

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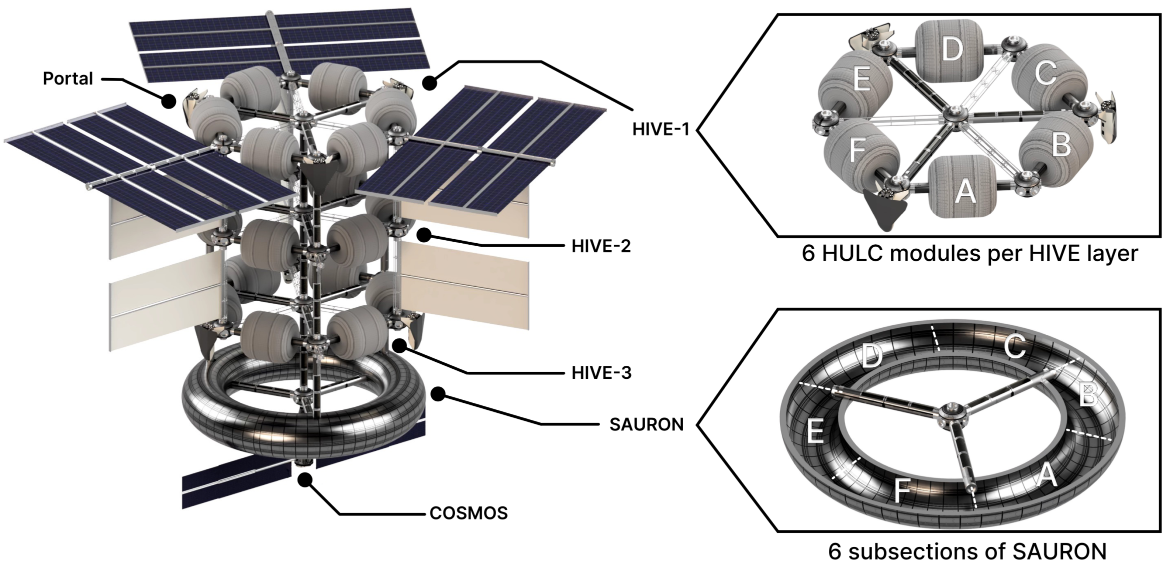
| | |
|---|---|
| $\alpha$ | angle of attack |
| $\bar{V}$ | volume coefficient |
| $\dot{q}$ | heat flux |
| $\epsilon$ | emissivity |
| $\Gamma$ | dihedral |
| $\gamma$ | ratio of specific heat capacities |
| $\Lambda$ | sweep |
| $\lambda$ | taper ratio |
| $\mathbf{x}$ | state vector For genetic algorithm |
| $\mu$ | dynamic viscosity |
| $\psi$ | flight path angle |
| $\rho$ | density |
| $\sigma$ | Stefan-Boltzmann constant |
| $\theta$ | surface inclination |
| $AR$ | aspect ratio |
| $b$ | wingspan |
| $BC$ | ballistic coefficient |
| $c$ | chord |
| $C_D$ | coefficient of drag |
| $C_L$ | coefficient of lift |
| $C_P$ | coefficient of pressure |
| $c_p$ | specific heat capacity at constant pressure |
| $D$ | drag |
| $g$ | gravitational acceleration |
| $h$ | height |
| $L$ | lift |
| $l$ | moment arm |
| $M$ | Mach number |
| $m$ | mass |
| $Pr$ | Prandtl number |
| $Q$ | heat load |
| $R$ | radius |
| $r$ | recovery factor |
| $Re$ | Reynolds number |
| $S$ | reference area |
| $s$ | downstream distance |
| $St$ | Stanton number |
| $T$ | temperature |
| $t$ | time |
| $V$ | velocity |
| $W$ | weight |
| $x$ | distance from stagnation point |

# Visual Overview




Structural Overview of the *Skyview*.

HULC sections.

| HULC | HIVE-1 | HIVE-2 | HIVE-3 |
|---|---|---|---|
| A | Rentable Lab | Crew Hab | Guest Hab |
| B | Rentable Lab | Crew Hab | Guest Hab |
| C | Permanent Lab | Crew Hab | Guest Hab |
| D | Permanent Lab | Communal Area | Communal Area |
| E | Lab Storage | Storage | Guest Hab |
| F | Lab Storage | Storage | Guest Hab |

SAURON sections.

| SAURON Section | Function |
|---|---|
| A | Reception & Atrium |
| B | Bar & Restaurant |
| C | Gym & Sauna |
| D | Multi-Sport Arena |
| E | Multi-Sport Arena |
| F | Storage |

## Glossary: Acronyms

Associated acronyms.

| Team | Acronym | Definition |
|---|---|---|
| ISH04 | BOND | Berthing Operational Network Device |
| ISH04 | COSMOS | Core Operations and Systems Module for Orbital Services |
| ISH04 | HAGRID | Hexagonal Assembly for General Resource Integration and Deployment |
| ISH04 | HIVE | Habitable Integrated Vessel |
| ISH04 | HULC | Habitational Unit and Life Capsule |
| ISH04 | NODE | Nodal Orbital Docking Extension |
| ISH05 | ARM | Autonomous Robotic Manipulator |
| ISH05 | ISRBS | Imperial Skyview Retractable Berthing System |
| ISH05 | ORA | Orbital Ring Assembly |
| ISH05 | SAURON | Space Assembled Utility Ring & Orbital Node |

# 1 Introduction

*Imperial Skyview Space Hotel* is a tourist oriented space station which provides guests the opportunity to stay for up to 7 days in low earth orbit as well as commercial access to research in micro-gravity environments. *Skyview* aims to have a capacity of 55 people (comprising 25 guests and 30 crew for guest services and commercial research), a lifespan of 25 years and to be operational by 2035. A Guest Transport Vehicle (GTV) is required to deliver passengers and cargo from the Earth's surface to *Skyview*. It also must be able to de-orbit from Skyview to bring passengers and cargo back to the Earth's surface. During the initial phase of re-entry through the atmosphere, the GTV will be travelling at near orbital velocity which will cause a high amount of heating and deceleration. A suitable Thermal Protection System (TPS) is required to protect the GTV from this heating.

The use of an existing GTV for *Skyview* would require an unfeasible number of vehicles over *Skyview's* lifespan. For example, SpaceX's *Crew Dragon* has a passenger capacity of 7, a lifespan of up to 15 flights, and a turnaround time of 5 months between flights [1]. To transport 25 passengers to and from *Skyview* every 7 days, approximately 330 *Crew Dragons* would be required over the lifespan of *Skyview*. Additional crew and cargo requirements further increase this number. The cost implications of such a fleet are discussed in [2], from which a decision was made by *ISH01: Project Management* to reduce the required number of GTVs.

*Portal* is a GTV designed uniquely for the needs of *Skyview*, aiming to decrease the number of GTVs required and provide greater luxury for its passengers. An ablative TPS (such as used on *Crew Dragon*) limits the lifespan of a GTV, as the TPS is consumed on each re-entry, therefore, an insulating TPS with highly reusable materials is essential. To ensure the maximum operating temperature of the insulating materials is not exceeded, a low ballistic coefficient is required [3], which is most easily achieved by increasing the area of the TPS. To reduce the g-loads experienced by passengers during re-entry, a high lift-to-drag design is necessary [3]. *Portal* achieves both a low ballistic coefficient and high lift-to-drag with a winged design as depicted in Fig. 1a. The winged design has the additional benefit of a glided landing on a runway - similar to conventional aircraft. When compared to a splash down on water, this reduces the impact on *Portal* - which is beneficial for passenger comfort and for limiting damage - as well as providing faster passenger disembarkment times and faster access to *Portal* for maintenance - reducing turnaround times. *Portal* will be transported to *Skyview* using ULA's *Vulcan-Centaur* [4]. This launch vehicle was chosen as it provided the largest fairing dimensions from the range of launch vehicles which had sufficient payload mass and TRL.

(a) Isometric view. (b) Inside fairing view.

Figure 1: 3D model of *Portal*.

To protect the TPS from debris during ascent, *Portal* must be able to fit inside *Vulcan's* fairing. To maintain a low ballistic coefficient whilst meeting this constraint, the wings are designed with a hinge mechanism to maximise wing area, shown in Fig. 1b. This was inspired by deployable re-entry vehicles such as NASA's *Adaptable Deployable Entry And Placement Technology (ADEPT)* [5] or

NASA's *Hypersonic Inflatable Aeroshell Demonstrator (HIAD)* [6]. Similar designs could not be implemented for *Portal* as at the time of writing their Technology Readiness Level (TRL) is too low. This is due to limited flight testing for larger payloads and the complex aerodynamic-structural coupling for inflatable and complicated mechanical mechanisms not being fully understood. A simple folding wing achieves a similar result - increasing the TPS area whilst meeting fairing constraints - and relies on higher TRL technologies. The simple hinge is simpler than the mechanism found on *ADEPT* and is similar to the canards on SpaceX's *Starship*, which have been successfully flight tested [7].

The proposed concept of *Portal* is similar to the *Space Shuttle*, which was a pioneer in the use of an insulating TPS and in high lift-to-drag designs but was severely flawed. It originally proposed a lifespan of 100 flights, though only achieved a lifespan of 39 flights due to these flaws [8]. These mainly consisted of; a single-stage-to-orbit design being inefficient; the TPS being damaged during ascent; issues in the re-use of first-stage engines; and material limitations. These flaws have been considered and mitigated in *Portal's* design by using a separate multi-stage launch vehicle and protecting the TPS during ascent. More recently, SpaceX's *Starship* has demonstrated that a concept based around an insulating TPS can be successful. Elon Musk (SpaceX's CEO) has an ambitious aim for *Starship* to be completely reusable with turnaround times as low as 1 hour [1] [7]. Using these examples as a baseline, *Portal* is specified to have a capacity of 8 passengers, a turnaround time of 1 month, and a lifespan of 100 flights. This reduces the required number of GTVs to 40 over the 25 year lifespan of *Skyview*.

This report outlines the sizing and trajectory optimisation of *Portal* to meet the heating constraints of the TPS material. The heat fluxes across the vehicle are also predicted for use in TPS design and mass estimations. The design of the TPS is detailed in [9]. The design of the propulsion system, the conceptual design of *Portal's* structures and estimates of its mass are detailed in [10]. The estimation and validation of aerodynamic forces on *Portal* at hypersonic speeds is detailed in [11].

# 2 Prerequisite Theory

To ensure the operating temperatures of the materials used for the TPS are not exceeded, the heat flux along the vehicle must be estimated. This requires knowledge of the re-entry vehicle's trajectory which requires knowledge of aerodynamic forces. The following sections all assume that the flow behaves as a continuum, which is valid for the majority of the trajectory, including at peak heating [12]. For heating of small-scale features to be estimated, direct Monte Carlo simulations are required.

## 2.1 Aerodynamics Forces

The pressure distribution $P$ along the surface determines the aerodynamic force on a body. This is usually modelled with a non-dimensional pressure coefficient $C_P$, defined as:

$$C_P = \frac{2}{\gamma M_\infty^2}\left(\frac{P}{P_\infty} - 1\right) \tag{1}$$

Where $P_\infty$ and $M_\infty$ are the free-stream pressure and Mach number respectively and $\gamma$ is the ratio of specific heats (taken as 1.4 for air throughout). For re-entry vehicles travelling at hypersonic velocities ($M_\infty > 5$), Newtonian theory can be used to readily estimate $C_P$ [3]. The theory assumes that the incoming fluid loses all momentum normal to the inclination angle $\theta$ of the surface it collides with. This allows for $C_P$ on exposed surfaces to be approximated as a function of $\theta$ through the following equation [3]:

[1]As of writing, these claims do not have significant evidence behind them.

$$C_P \approx 2\sin^2\theta \quad (2) \qquad\qquad C_P \approx C_{P_{max}}\sin^2\theta \quad (3)$$

For all other surfaces not directly exposed to the free stream, an approximation of $C_P \approx 0$ is made [3]. Modified Newtonian theory adjusts the constant term to include the effects of stagnation pressure loss $^{P_{os}}/_{P_o}$ due to the incoming fluid passing through a normal shock (which adds a dependency on $M_\infty$), improving the accuracy of results for blunt bodies which produce a curved bow shock [3]. The ratio of stagnation pressure loss is calculated through the Rayleigh-Pitot formula, shown below [3], which is then used to calculate $C_{P_{max}}$ with Eq. (1).

$$\frac{P_{os}}{P_\infty} = \left[\frac{(\gamma+1)^2 M_\infty^2}{4\gamma M_\infty^2 - 2(\gamma-1)}\right]^{\frac{\gamma}{\gamma-1}} \left[\frac{2\gamma M_\infty^2 - (\gamma-1)}{\gamma+1}\right] \tag{4}$$

Note, in the limit of $M_\infty \to \infty$ and $\gamma \to 1$ (which is used to approximate real-gas effects [3]) then $C_{P_{max}} \to 2$, the two methods converge. So, the difference between methods is only appreciable for low Mach numbers.

## 2.2 Re-Entry Trajectory

A 2-dimensional trajectory of a re-entry vehicle only considers the downstream distance travelled $s$ and the height above the Earth's surface $h$ [12]. This will provide the necessary values for heat estimation but neglects any effects of bank angle and longitudinal/latitudinal position. A re-entry vehicle can be modelled as a point mass which travels over a spherical non-rotating Earth. The main forces acting on the re-entry vehicle are its weight due to mass and gravitational acceleration $W = mg$, lift $L$, and drag $D$. The orientation of these forces is determined by the flight path angle $\psi$ between the free stream velocity $V_\infty$ and the local horizon (defined as positive for an upward inclination). The lift and drag forces are modelled using the non-dimensional coefficients $C_L$ and $C_D$, which are scaled by the free stream dynamic pressure $^1/_2\rho_\infty V_\infty^2$ (where $\rho$ is density) and reference area $S_{ref}$, as follows [13]:

$$L = \frac{1}{2}\rho_\infty V_\infty^2 S_{ref} C_L \quad (5) \qquad\qquad D = \frac{1}{2}\rho_\infty V_\infty^2 S_{ref} C_D \quad (6)$$

Note, the values taken for $S_{ref}$ depend on the method used to evaluate $C_L$ or $C_D$. Unless stated otherwise, $S_{ref}$ was taken as the projected planform area of the entire vehicle. For initial estimates, a re-entry vehicle can be estimated as a flat plate at an angle of attack $\alpha$, for which the following results from modified Newtonian theory can be applied [3].

$$C_L \approx C_{P_{max}}\sin^2\alpha\cos\alpha \quad (7) \qquad\qquad C_D \approx C_{P_{max}}\sin^3\alpha \quad (8)$$

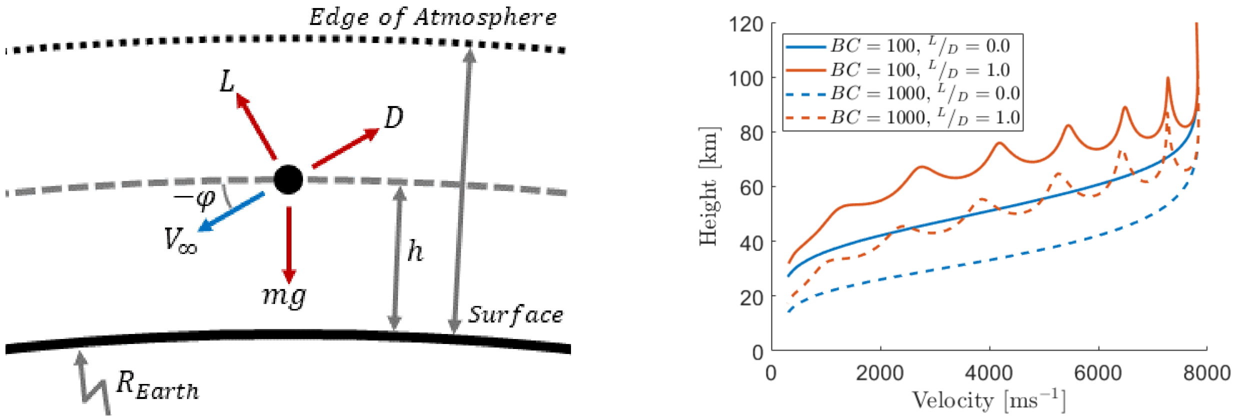
The lift-to-drag ratio $^L/_D$ alongside the ballistic coefficient $BC$ are the key parameters for a re-entry vehicle. The ballistic coefficient provides a measure of the mass of a vehicle compared to its drag and is defined as [3]:

$$BC = \frac{m}{C_D S_{ref}} \tag{9}$$

A vehicle with a low $BC$ will experience greater deceleration, leading to reduced velocities earlier in re-entry and thus lower heat fluxes and heat load [14]. This is why it was desirable to maximise the planform area of *Portal* - it reduced the $BC$. It also explains why the number of passengers was not greatly increased from existing vehicles - it would increase the mass and thus increase the $BC$.

By analysing the free-body diagram shown in Fig. 2a, applying Newton's 2nd law in-line and normal to the flight path, accounting for the rotating reference frame, then substituting in Eq. (5), Eq. (6) and Eq. (9), the following equations of motion are derived [12], where $R_E$ is the radius of the Earth and $t$ is time:

$$\frac{dV_\infty}{dt} = -\frac{\rho_\infty V_\infty^2}{2}\frac{1}{BC} - g\sin\psi \qquad (10) \qquad V_\infty \frac{d\psi}{dt} = \frac{1}{2}\rho_\infty V_\infty^2 \frac{L}{D}\frac{1}{BC} - \left(g - \frac{V_\infty^2}{R_E+h}\right)\cos\psi \quad (11)$$




(a) Free-body diagram of a re-entry vehicle.

(b) Example trajectories on a V-h diagram.

Figure 2: Free-body diagram and example re-entry trajectories.

With the inclusion of an atmospheric model to estimate the free-stream conditions at a given altitude, $T_\infty, \rho_\infty, P_\infty = f(h)$, and the initial conditions $V_\infty(t_o)$, $h(t_o)$, and $\psi(t_o)$, the trajectories can be found with an appropriate numerical scheme for Eq. (10) and Eq. (11). The choices for which are summarised in Tab. 1. A brief time-step study concluded that a time-step of $\Delta t \leq 3s$ produced an error of less than 1 % when compared to the converged value. Full results of the study can be seen in Appendix A.

Table 1: Setup for trajectory simulations.

| Atmospheric Model | Numerical Scheme | $V(t_o)$ [ms$^{-1}$] | $h(t_o)$ [km] | $\psi(t_o)$ [°] | $\Delta t$ [s] |
|---|---|---|---|---|---|
| NRLMSISE-00 [15] | Explicit Euler | 7800 [16] | 120 [16] | -1.0 [16] | 3.0 |

The results of a trajectory analysis are commonly presented in velocity-height (V-h) diagrams. Example results using the aforementioned methodology are shown in Fig. 2b. An important point is the identification of "skips" caused by a high $L/D$, which is of importance to the trajectory optimisation in Section 4. As the vehicle descends into the atmosphere the density increases, which causes an increase in lift through Eq. (5), which for vehicles with a high $L/D$ will induce a climb. Eventually, the vehicle reaches an altitude with a lower density that reduces the lift to a point where descent begins again. This process repeats, causing the "*skips*".

## 2.3 Heating

To estimate the conductive heat flux $\dot{q}_{cond}$ from the air in the boundary layer to the TPS across the vehicle one of three empirical equations are used depending on the component geometry. All of these relations rely on an estimate of the local wall temperature $T_w$ and the adiabatic wall temperature $T_{aw}$. A recovery factor $r$ describes the ratio of kinetic energy recovered by air as it decelerates from free-stream conditions to stagnation conditions at the wall and can be used to estimate $T_{aw}$ as follows:

$$T_{aw} = r(T_o - T_e) + T_e \qquad (12)$$

Where $T_o$ is the stagnation temperature and $T_e$ is the temperature at the edge of the boundary layer. The recovery factor can be taken as $r_{lam} = \sqrt{Pr}$ for laminar flows and as $r_{turb} = \sqrt[3]{Pr}$ for turbulent flows ($Pr = 071$ is the Prandtl number for air) [3]. The prediction of where transition occurs is highly complex, so to simplify the process a crude approximation for the Reynolds number at transition of $Re_{trans} \approx 10^6$ [12] was used throughout.

The wall temperature is more difficult to calculate as it requires a coupled solution between the heat flux to the TPS and the heat conduction through the TPS. Wherever $T_w$ appears a conservative estimate of $T_w \approx 200$ K is made in the initial stages - which will overestimate heat fluxes. For the final sizing of the TPS, the coupled problem was solved, as detailed in [9].

Since materials specify a maximum operating temperature $T_{max}$, it can be useful to convert this to an equivelant maximum heat flux. This can be done by assuming radiative equilibrium, as shown in the following equation where $\sigma = 5.67 \times 10^{-8}$ is the Stefan–Boltzmann constant and $\epsilon$ is the emissivity of the material:

$$\dot{q}_{max} = \sigma \epsilon {T_{max}}^4 \tag{13}$$

By considering the material choice for the nose of *Portal*, which was Reinforced Carbon-Carbon Composite (reasoning detailed in [9]), for which $T_{max} = 1920$ K [17] and $\epsilon = 0.78$ [17], a heat flux limit of $\dot{q}_{max} = 600$ kWm$^2$ using Eq. (13) was found. The heat load is also defined as $Q = \int \dot{q} dt$, and is useful in determining the mass of the TPS.

### 2.3.1 Heating of a Sphere

The stagnation point heating behind a curved shock on a sphere of radius $R_N$ can be estimated through the following empirical equation [12]:

$$\dot{q}_{cond} = 1.83 \times 10^{-4} \sqrt{\frac{\rho_\infty}{R_N}} V_\infty^3 \left[1 - \frac{T_w}{T_{aw}}\right] \tag{14}$$

This was used to approximate the heating on the nose of *Portal* and shows that a blunter nose will reduce heating. This provides motivation in Section 3.1 for maximising the nose radius of the vehicle.

### 2.3.2 Heating of a Swept Cylinder

The stagnation point behind a curved shock on a cylinder of radius $R_{l.e.}$ and sweep angle $\Lambda_{l.e.}$ can be estimated with the following empirical adjustments to Eq. (14) for a sphere of equivalent radius [18]:

$$\dot{q}_{cond} = \frac{\dot{q}_{sphere}}{\sqrt{2}} \cos^{1.1} \Lambda_{l.e.} \tag{15}$$

This was used to approximate the heating on the leading edge of the wing and shows that the heating here can be of similar magnitude to the heating on the stagnation point. This must be considered during the design of the wing in Section 3.2.

### 2.3.3 Heating across a Flat Plate

For a flat plate with approximately zero pressure gradient, the reference temperature method can be applied [19], which predicts the Stanton number $St$. This can be related to the heat flux on the vehicle through its definition [3]:

$$St = \frac{\dot{q_w}}{\rho_e V_e c_p T_{aw} \left[1 - \frac{T_w}{T_{aw}}\right]} \tag{16}$$

Where $\rho_e$ and $V_e$ are the respective properties evaluated at the edge of the boundary layer and $c_p$ is the specific heat capacity at constant pressure - not to be confused with the coefficient of pressure $C_P$. Reynold's Analogy relates the Stanton number to the skin friction coefficient $C_f$ (for which many in-compressible relations already exist) through $St \approx {}^{C_f}/_2$. The in-compressible relations can be used to evaluate $St$ in a compressible boundary layer if a reference temperature $T^*$ which lies between $T_e$ and $T_w$ is used [19].

To find the properties along the boundary layer edge, it is assumed that the flow passes through a nearly normal shock and is then expanded around the body by a series of isentropic expansion waves. This is a relatively accurate description of the flow for blunt bodies. Thus, normal shock equations followed by isentropic relations can be applied. The values downstream of the normal shock ($M_s$, $P_s$, $\rho_s$, $T_s$) are found by:

$$\begin{cases} M_s^2 = \frac{2+(\gamma+1)M_\infty^2}{2\gamma M_\infty^2-(\gamma-1)} & \frac{P_s}{P_\infty} = 1 + \frac{2\gamma}{\gamma+1}\left(M_\infty^2 - 1\right) \\ \frac{\rho_s}{\rho_\infty} = \frac{(\gamma+1)M_\infty^2}{2+(\gamma-1)} & \frac{T_s}{T_\infty} = \frac{P_s}{P_\infty}\frac{\rho_\infty}{\rho_s} \end{cases} \tag{17}$$

The pressure along the edge of the boundary layer $P_e$ was found using modified Newtonian theory, Eq. (3), which then allowed for all other boundary layer edge properties $T_e$, $\rho_e$, and $M_e$ to be found through the following isentropic relations [13]:

$$\frac{P_e}{P_s} = \left(\frac{\rho_e}{\rho_s}\right)^\gamma = \left(\frac{T_e}{T_s}\right)^{\frac{\gamma}{\gamma-1}} \tag{18}$$

$$M_e^2 = \frac{2}{\gamma-1}\left[\frac{T_e}{T_s}\left(1+\frac{\gamma-1}{2}M_s^2\right)-1\right] \tag{19}$$

$T^*$ can then be calculated, depending on the flow condition, and then the associated viscosity through Sutherland's Law as shown below [19]:

$$\begin{cases} \left(\frac{T^*}{T_e}\right)_{lam} = 0.45 + 0.55\frac{T_w}{T_e} + 0.16r\frac{(\gamma-1)}{2}M_e^2 \\ \left(\frac{T^*}{T_e}\right)_{turb} = 0.5 + 0.5\frac{T_w}{T_e} + 0.16r\frac{(\gamma-1)}{2}M_e^2 \end{cases} \tag{20}$$

$$\mu^* = 1.458 \times 10^{-6}\frac{T^{*\frac{3}{2}}}{T^* + 110.4} \tag{21}$$

The associated density $\rho^*$ is calculated using the ideal gas law and by assuming the pressure is constant throughout the boundary layer $\rho^* = \rho_e {T_e}/{T^*}$. The Reynolds number $Re_x^*$ at distance $x$ from the stagnation point on the body can then be evaluated, and thus $St$ at that point [3]:

$$Re_x^* = \frac{\rho^* V_e x}{\mu^*} \tag{22}$$

$$\begin{cases} St_{lam}^* = \frac{0.332}{\sqrt{Re_x^*}}Pr^{-2/3} \\ St_{turb}^* = \frac{0.0296}{\sqrt[5]{Re_x^*}}Pr^{-2/3} \end{cases} \tag{23}$$

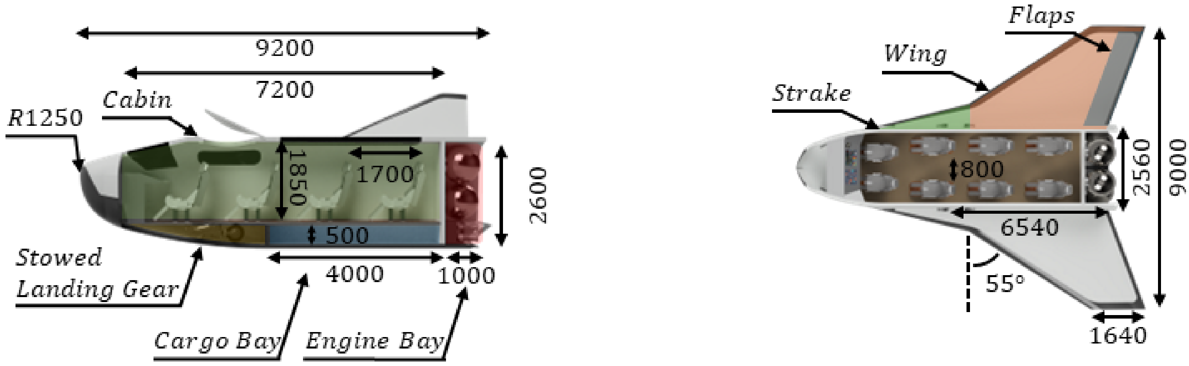
The heating at the evaluated point can then be calculated through Eq. (16).

# 3 The Sizing of *Portal*

The design of *Portal's* geometry considered many factors to size the fuselage, wings and stabilisers. The key dimensions of the final design are highlighted in Fig. 3. Additional views of *Portal* can be seen in Appendix B.

## 3.1 Fuselage Sizing

The cabin was sized to fit 8 passenger seats using historical data from luxurious business jets as a baseline [20]. The seat pitch was increased from the baseline value to allow room for rotation of the seats, allowing the passengers to be oriented into a position for improved g-load tolerance during launch and re-entry [21]. The aisle width was also increased to allow for easier access to the underfloor cargo bay. Every seat has at least one window view which provides a unique view of *Skyview* before docking and of the upper atmosphere during re-entry. The docking port, where the passengers will embark/disembark and where *Portal* will connect to the *NODE* module on *Skyview*, was placed on the roof of the cabin. Underneath the floor of the cabin is the cargo bay, which has a volume of 5 m$^3$ (specification set by *ISH03: Operation, Interior Design & Guest Experience* [22]) for




(a) Side view.

(b) Top view.

Figure 3: Views of a 3D model for *Portal* (dimensions in mm).

the passengers' personal effects and a small amount of station consumables. The seats in the cabin can be removed to provide up to 30 m$^3$ of extra cargo volume for missions with extra requirements. The front landing gear is stowed in front of the cargo bay and the rear landing gear are stowed in the wing. The dimensions of the landing gear were taken from similarly sized jet aircraft as an initial estimate [20]. Behind the cabin is the engine bay which houses the propulsion systems for emergency abort and de-orbit burns. The sizing of these systems is detailed in [10].

The exterior geometry of the fuselage was designed to house the previously mentioned compartments whilst maximising the nose radius of the vehicle. Requirements for pilot field-of-vision were not considered as autonomous landing was to be implemented for *Portal*. By using a self-developed *MATLAB* tool to adjust the configuration of the GTV and through several iterations of manual design, a nose radius of $R_N = 1.25$m was achieved for the final design. This was then converted into a 3D model in *Fusion-360* for use by other teams.

## 3.2 Wing Sizing

The wing shape was based on findings from the *Space Shuttle* development programme, which found the parameters which minimised wing weight when accounting for the TPS weight [23]. The key parameters are aspect ratio ($AR = {}^{b^2}/_{S_{ref}}$, the ratio of wing span squared to wing area), taper ratio ($\lambda = {}^{c_{tip}}/_{c_{root}}$, the ratio of tip to root chord) and leading edge sweep ($\Lambda_{l.e.}$). Slight alterations were made to these values to ensure *Portal* fits inside *Vulcan's* fairing, to increase subsonic performance, and to increase the minimum radius along the leading edge. The chosen parameters are shown in Tab. 2 alongside comparative values. A dihedral of $\Gamma = 5°$ was added to ensure lateral stability at subsonic velocities [20].

Table 2: Key geometric parameters of the wing alongside comparative values.

| Vehicle | $AR$ | $\lambda$ | $\Lambda$ [°] |
|---|---|---|---|
| Suggested Values [23] | 2.5 | 0.2 | 60 |
| *Portal's* Wing | 2.2 | 0.25 | 55 |

With a known shape, the wing size is then determined by wing loading ${}^{W}/_{S_{ref}}$, (where $S_{ref}$ is the projected planform area of the wing), which has a close correlation to $BC$. By using an initial mass estimate of $m \approx 15000$ kg (reasoning detailed in [10]), assuming that the fuselage accounts for 30 % of the total area and assuming a constant $\alpha$, the $BC$ and ${}^{L}/_{D}$ can be found by using Eq. (7) and (8) for a given ${}^{W}/_{S_{ref}}$. The trajectory, and therefore the max heat flux across the re-entry vehicle can then be found. The resultant $BC$, ${}^{L}/_{D}$ and associated $\dot{q}_{max}$ for a range of wing loadings is shown in Fig. 4. This showed that a lower wing loading reduced max heating - which is to be expected

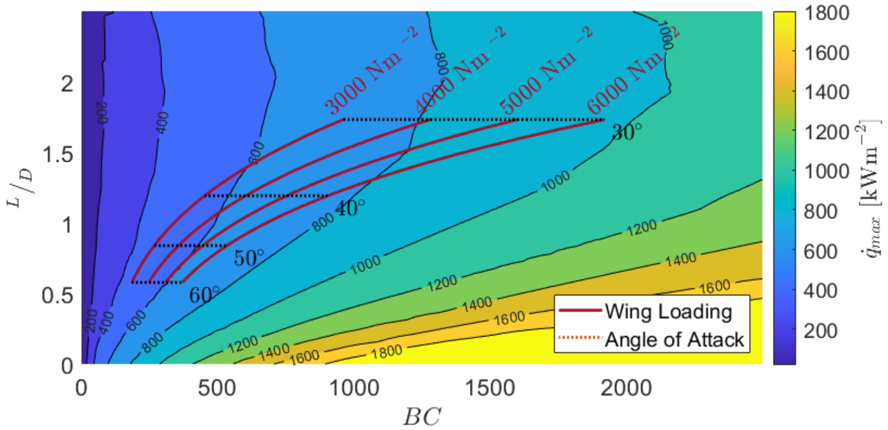



Figure 4: Carpet Plot of wing-loading against angle of attack.

due to the correlation with ballistic coefficient - thus a low as possible wing loading is desirable. An analysis showed that the lowest wing loading which ensured that *Portal* would fit in *Vulcan's* fairing was 4000 Nm$^{-2}$ (analysis covered in [10]). Such a Wing-loading met the $\dot{q}_{max} < 600$ kWm$^2$ constrain for $\alpha > 44°$, so was selected. This resulted in a wing area of 36.8 m$^2$ and an overall area of 45.8 m$^2$ for the geometry depicted in Fig. 3b.

A *NACA-2415* aerofoil profile was chosen for the main wing due to its' smooth lower surface which helps minimise wing-body interference heating [12]. The relatively high thickness also increases the radius at the leading edge (resulting in $R_{le_{min}} \approx 0.20$ m) and ensures the heating on the leading edge is lower than on the nose. A higher thickness aerofoil could not be used, as it significantly decreases sub-sonic performance [20].

The wing must also provide sufficient performance during landing. A constraint for touchdown velocity of $V_{TD} \leq 150$ knots and an actual landing distance of $ALD \leq 4000$ m at sea level was applied, which were the same constraints applied during the development of the *Space Shuttle* [23]. The touchdown velocity is taken as 1.1 times the stall velocity in accordance with FAR-25 [24] and is calculated via a simple re-arrangement of Eq. (5) and an empirical relation is used to estimate the $ALD$[20], both are shown below:

$$V_{TD} = 1.1\sqrt{\frac{2W/S_{ref}}{\rho C_{L_{max}}}} \quad (24) \qquad ALD = 0.51\frac{W/S_{ref}}{C_{L_{max}}} + s_a \quad (25)$$

Where $s_a$ is the approach distance before touchdown which was taken as 305 m for a 3° glide approach [20]. Both equations require an estimate of the maximum coefficient of lift $C_{L_{max}}$. For a low aspect ratio wing ($AR \leq 4$), the following equation can be used [20]:

$$C_{L_{max}} = (C_{L_{max}})_{base} + \Delta C_{L_{max}} \quad (26)$$

Where $(C_{L_{max}})_{base} \approx 0.91$ and $\Delta C_{L_{max}} \approx 0$ for *Portal's* wing geometry. The values are found from graphical data [20], which are presented in Appendix C. The data shows a general trend that increases in thickness and leading edge sweep decrease both $C_{L_{max}}$. High-lift devices were found to be necessary to meet the landing requirements. Simple plain flaps were used due to the low complexity which provides easier integration with the TPS. These flaps increase the sectional lift coefficient of the aerofoil by $\Delta C_{l_{max}} = 0.9$ [25] and extend from 40 % to 95 % span to maximise the flapped area of the wing. The increase in lift coefficient due to the flaps can be estimated as [20]:

$$\Delta C_{L_{max}} = 0.9\Delta C_{l_{max}}\frac{S_{flapped}}{S_{ref}}\cos\Lambda_{hinge} \quad (27)$$

The inclusion of a strake can further increase $C_{L_{max}}$ by the creation of a vortex streak which delays separation at high angles of attack. Wind tunnel studies for a similar geometry wing showed this can increase $C_{L_{max}}$ by up to 15 % [26]. After evaluating all the above equations, a value of $C_{L_{max}} = 1.39$ was found. Eq. (24) and (25) were then evaluated resulting in $V_{TD} = 147$ knots and $ALD = 1780$ m which met the required constraints.

### 3.3 Stabiliser Sizing

To ensure subsonic stability of *Portal*, an empennage was required. A V-tail configuration was chosen as it reduced the vertical dimensions of the stabilisers and reduced interference with *Skyview* during docking. The key geometric parameters for the stabiliser were based on historical data for fighter jets [20] which have a similar configuration to *Portal.* The chosen parameters are shown in Tab. 3.

Table 3: Key geometric parameters of the stabiliser alongside comparative values.

| Vehicle | $AR$ | $\lambda$ | $\Lambda$ [°] |
|---|---|---|---|
| Suggested Values [20] | $0.6 \rightarrow 3$ | $0.2 \rightarrow 0.4$ | $\geq \Lambda_{wing}$ |
| *Portal's* Stabiliser | 1.0 | 0.2 | 55 |

To size the stabilisers, an equivalent horizontal and vertical stabiliser are sized using tail volume coefficients, then the required V-tail size can be found through a *NACA* finding that showed conventional and V-tail stabilisers require the same wetted area [20]. The tail volume coefficients for vertical and horizontal stabilisers, $\bar{V}_{HT}$ and $\bar{V}_{VT}$ respectively, take into account the wing geometry (using either mean aerodynamic chord $\bar{c}$ or wing span $b$, and wing area $S_{ref}$) and the moment arm of the stabilisers $l$ (the distance between the aerodynamic centre of the aircraft and the aerodynamic centre of the stabiliser). They can be used to size the area of a horizontal and vertical stabiliser, and thus an equivalent V-tail, as follows [20]:

$$S_{HT} = \frac{\bar{V}_{HT} S_{ref} \bar{c}}{l} \quad (28) \qquad S_{VT} = \frac{\bar{V}_{VT} S_{ref} b}{l} \quad (29) \qquad S_{V-tail} = \frac{S_{HT} + S_{VT}}{2} \quad (30)$$

For an initial estimate, $l$ was taken as 50 % of the overall vehicle length and values of $\bar{V}_{VT} \approx 0.07$ and $\bar{V}_{HT} \approx 0.40$ were used based off historical data for fighter jets [20]. The required dihedral $\Gamma$ can be found by a force balance of the equivalent horizontal and vertical stabilisers [20], resulting in:

$$\Gamma_{V-tail} = \tan^{-1}\sqrt{\frac{S_{VT}}{S_{HT}}} \quad (31)$$

The resultant V-tail has an area of $S_{V-tail} = 5.05m^{\ 2}$ per stabiliser and a dihedral of $\Gamma = 41°$.

## 4 Trajectory Optimisation

During re-entry, *Portal* can vary its angle of attack through the range $15° \leq \alpha \leq 60°$. The lower limit cannot be exceeded as otherwise the upper surface (where there is a lower level TPS) is exposed to high heating and the upper limit cannot be exceeded due to control surface limitations [23]. The aerodynamic coefficients for *Portal* across the specified angle of attack range and a range of Mach number are obtained in [11] by applying modified Newtonian theory, Eq. (3), to a 2-dimensional discretised model of the geometry depicted in Fig. 3a. Fig. 5 shows the trajectories for constant angles of attack, showing the minimum and maximum values which met the applied constraints - which are explained in due course.

The high lift-to-drag results in a "*skipping*" behaviour, as noted in Section 2.2. This behaviour was determined to be undesirable with a belief that it adds to the flight time duration which increases

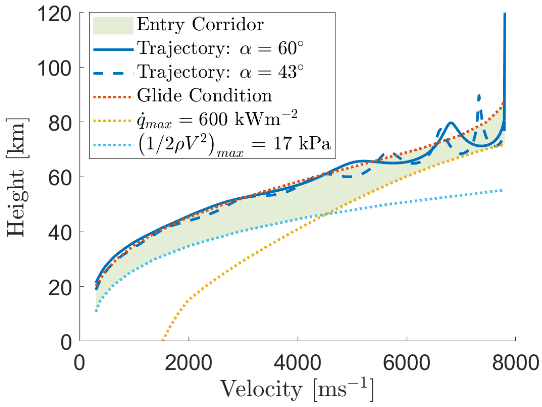



Figure 5: V-h diagram for *Portal* at constant angles of attack.

the heat load on the vehicle - which is undesirable due to the associated increase in TPS mass as explained in [10]. A solution to minimise the skipping behaviour is to vary the angle of attack in time ($\alpha = f(t)$) to achieve a desired trajectory on a V-h diagram.

### 4.1 Desired Trajectory

Before choosing a desired trajectory, the range of possible trajectories which form an entry corridor is considered. The $\dot{q}_{max}$ constraint has previously been introduced in Section 3.2. Since stagnation point heating, Eq. (14), is a function of $V_\infty$ and $\rho_\infty$, where the atmospheric model provides $\rho_\infty = f(h)$, the $\dot{q}_{max}$ constraint can be plotted as a curve on the V-h diagram for a given $R_N$ as demonstrated in Fig. 5. A load factor constraint is also applied in the form of a max dynamic pressure $\left({}^1\!/_2\rho_\infty V_\infty^2\right)_{max}$ to ensure that *Portal* stays within set structural limitations. For a chosen load factor of 3 (in line with similar vehicles [23]) then $L_{max} = 3W$ which through Eq. (5) results in $\left({}^1\!/_2\rho_\infty V\infty^2\right)_{max} = 17$ kPa for *Portal*. This constraint can also be plotted on the V-h diagram as demonstrated in Fig. 5. To ensure passenger safety, a g-load constraint is applied in line with NASA standards [21]. This specifies how long a certain g-load in a given orientation can be tolerated before ill effects are experienced. The most constraining values are applied, with seat position being adjusted to ensure passenger comfort. The limits are as follows:

$$\begin{cases} \frac{dV_\infty}{dt} \leq 12g & \text{for } t \leq 0.04s \\ \frac{dV_\infty}{dt} \leq 5g & \text{for } t \leq 0.1s \\ \frac{dV_\infty}{dt} \leq 3g & \text{for } t \leq 180s \end{cases} \tag{32}$$

An upper limit on the V-h diagram can be taken as the gliding trajectory. This is found by assuming ${}^{d\psi}\!/_{dt} \approx \psi \approx 0$, for which Eq. (11) becomes an algebraic equation. The velocity required to maintain horizontal steady level flight at a given altitude can then be found for *Portal*'s $BC$ and ${}^L\!/_D$ for where $C_{L_{max}}$ occurs.

The glide limit alongside max heating and dynamic pressure constraints form the entry corridor, highlighted in Fig. 5. This is the area in which all suitable trajectories are believed to lie. The glide limit was taken to be the desired trajectory, as it was believed to minimise heat flux and load on the vehicle.

### 4.2 Genetic Algorithms

To find a suitable function $\alpha = f(t)$ which achieves the desired trajectory, genetic algorithms were employed. This is a computational method inspired by natural selection. An initial population is generated and then scored using a cost function, the best-performing individuals from that population are retained and combined in different manners to create a new population. The process repeats until the change in performance between generations falls below a given tolerance [27]. This method of optimisation was selected as the use of *MATLAB's Global Optimisation Toolbox* allowed for quick setup and flexible implementation [27]. However, it is a very computationally expensive method and is unlikely to find the global minima for a given problem.

For implementation of the genetic algorithm, the function $\alpha = f(t)$ must be represented as a 1-D vector. This was done by specifying $\alpha = f(t)$ to be a piecewise linear function which holds *Portal* at a constant angle of attack for set periods of time, similar to the flight profile used for the *Space Shuttle* [23]. The control points for this function are specified in vector form as follows:

$$\mathbf{x} = \left[\alpha_1, t_1, ..., \alpha_N, t_N\right] \tag{33}$$

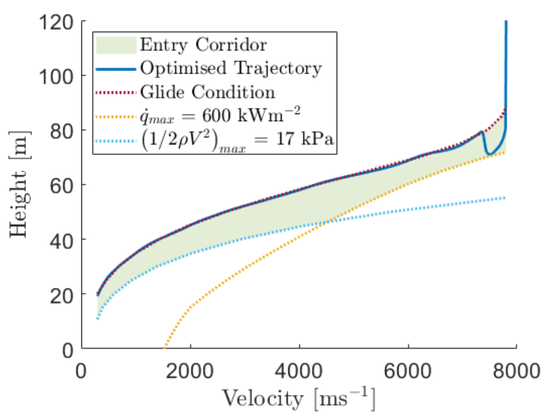
Linear constraints were applied to $\mathbf{x}$ to ensure the angle of attack constraints were not violated, that the points increased in time and to limit the rate of turn to 1° per second. The cost function on which each individual is evaluated was chosen to be the area between the trajectory achieved from a series of control points $h(\mathbf{x})$ and the desired trajectory $h_{desired}$. This is formulated mathematically as:

$$Cost = \int_0^{V(t_o)} |h(\mathbf{x}) - h_{desired}| dV_\infty \tag{34}$$

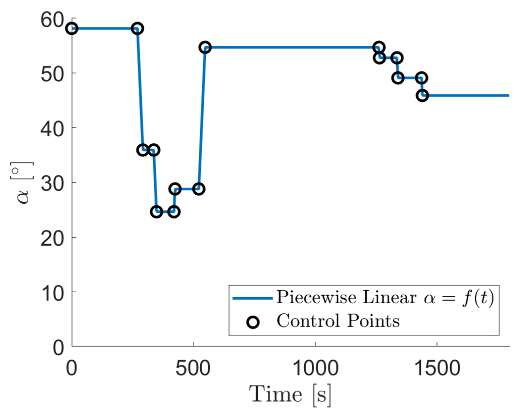
Where the integral is numerically evaluated using the trapezoidal method. This cost function was chosen due to previous studies showing its effectiveness [28]. Non-linear constraints were applied to enforce that the resultant trajectory does not violate the $\dot{q}_{max}$, $\left(1/2\rho_\infty V_\infty^2\right)_{max}$ or g-load constraints. This is carried out by adding a large penalty to the cost function whenever they are violated. The hyper-parameters of the genetic algorithm were chosen using *MATLAB's* recommended settings for similar optimisation problems [27], with a max of 250 generations and a population size of 50 per generation. Increasing these values is preferable but were limited by the computational power available.

## 4.3 Results

The resultant trajectory from this setup is presented in Fig. 6a and the associated angle of attack profile is presented in Fig. 6b. Except for an initial dip, the obtained trajectory matches the glide limit very closely, with a mean absolute error between the two profiles of 1.03 %. However, the acheiced trajectory does not have as clear benefits as initially believed.




(a) Optimised trajectory in a V-h diagram.




(b) Optimised piecewise linear function of angle of attack in time set by control points.

Figure 6: Results for the optimised trajectory.

Tab. 4 shows key numerical values along a range of trajectories. A key point to be made is that the difference in values between the optimised trajectory and a trajectory for $\alpha = 50°$ is small.

Furthermore, the average $\alpha$ across the optimised trajectory is $\bar{f}(t) = 49.4°$, which implies that variance in the angle of attack does not necessarily have a large effect on the key values during re-entry. The importance of $BC$ is further highlighted, as the maximum angle of attack trajectory was the one to minimise heat flux and load. However, such a trajectory was not chosen due to the risk of staying close to the limit of the vehicle for an extended period. Another interesting point is that the heat flux was the most constraining factor across all valid trajectories.

Table 4: Key numerical values for a range of possible trajectories for *Portal*.

| $\alpha$ [°] | $t_{end}$ [minutes] | $s_{end}$ [km] | $\dot{q}_{max}$ [kWm$^{-2}$] | $Q$ [MJm$^{-2}$] | $\left(1/2\rho_\infty V_\infty^2\right)_{max}$ [kPa] | $dV_\infty/dt_{max}$ [g] |
|---|---|---|---|---|---|---|
| 43 | 37.0 | 11,000 | 597 | 476 | 6.03 | 0.87 |
| 50 | 30.0 | 8,700 | 563 | 354 | 5.00 | 1.08 |
| 60 | 22.6 | 6,200 | 548 | 245 | 4.11 | 1.41 |
| $f(t)$ | 30.1 | 8,800 | 570 | 379 | 5.56 | 1.15 |

The heat fluxes and accompanying parameters were evaluated across the body for the optimised trajectory, which was then used for the design of the TPS, detailed in [9], and for initial estimates for the mass of *Portal*, detailed in [10]. The heat flux values are provided in Appendix D.

## 5 Further Work

This report, alongside [9, 10, 11] has detailed the initial design of *Portal* and have provided a baseline that proves a feasible concept. There is much further work to be done for the design to be considered complete, which extends far beyond the scope of the project. Further work which would expand on the contents of this report could include:

- Implementation of a less simplistic trajectory model. Accounting for bank angle, longitudinal/latitudinal location, the effects of the Earth's rotation and the wind in its atmosphere. Furthermore, work on the cross-range of *Portal* to reach a target destination could be carried out with such a model.
- Consideration of aerodynamic forces during supersonic, transonic and subsonic phases. Though the landing phase has been considered in this report, there is still much work to be done to ensure stability and desirable handling characteristics throughout all regimes of flight. However, the impact on the geometry of *Portal* would be minimal, as the constraining factors of flight are the hypersonic parts of initial re-entry and landing.
- Investigation into the optimisation of the de-orbit path of *Portal*. This would have an effect on the initial velocity and flight path angle of the re-entry trajectory and may have a greater effect on the heating of the vehicle than varying angle of attack during re-entry.

## 6 Conclusions

Thus, *Portal's* fuselage has been sized to meet requirements for passenger comfort, accommodation of other systems, and to minimise heating. The wing has been sized to minimise heating during re-entry whilst ensuring landing constraints are met and that *Portal* can fit inside *Vulcan's* fairing. Stabilisers have also been sized to provide subsonic stability during the landing phase. A genetic algorithm was used to find an angle of attack profile to follow the glide limit and mitigate skipping behaviour. However, the benefits of such a trajectory were found to be minimal, and that a trajectory with an equivalent average angle of attack provides similar values for range, g-load, dynamic pressure and heating despite skipping behaviour.

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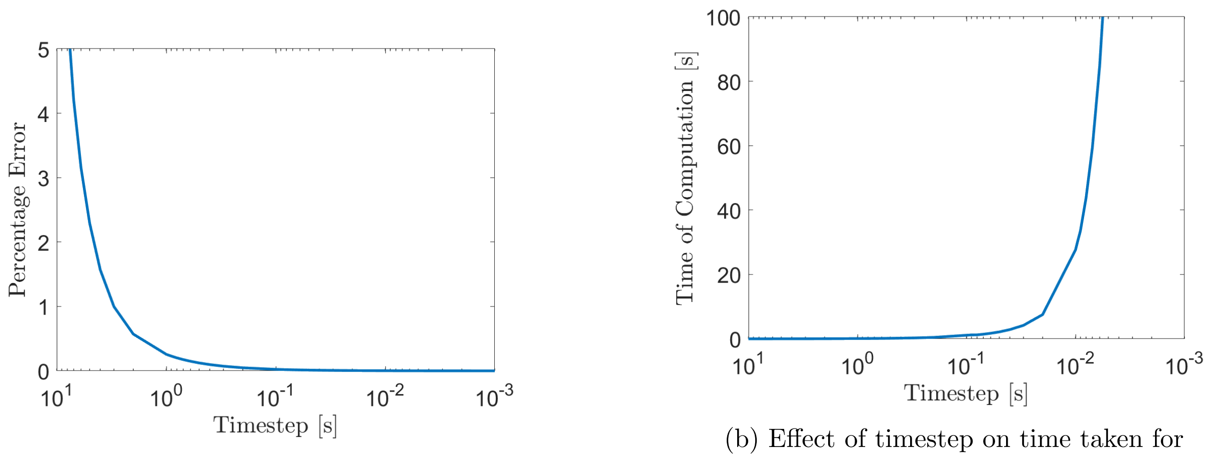
# Appendices

## A Trajectory Timestep Study

For a $BC = 300$ and $^L/_D = 0.8$ - which are similar to the range of values expected for *Portal* - a timestep study was carried out. Results are shown in Fig. 7.




(a) Effect of timestep on relative error.

(b) Effect of timestep on time taken for computation.

Figure 7: Effects of timestep on performance of trajectory simulations.

## B Additional Views

Figure 8: Views of a 3D model for *Portal* in landing configuration.

Figure 9: View of a 3D model for *Portal* in folded configuration.

Figure 10: View of a 3D model for *Portal* in docked configuration.

# C Graphical Data for Low Aspect Ratio Wings

To evaluate $C_{L_{max}}$, the correction factors $C1$ and $C2$ are read from Fig. 11 for a given taper ratio $\lambda$. A correction for Mach number, $\beta = \sqrt{1 - M_\infty^2}$, is applied, which for the case of landing is $\beta = 1$. $(C_{L_{max}})_{max}$ can then be read off from Fig. 12 and $\Delta C_{L_{max}}$ can be read off from Fig. 13 using $\Delta y = 26 t/c$ for a NACA 4-digit aerofoil. Note, in these figures $A$ is aspect ratio. Eq. (26) can then be evaluated.

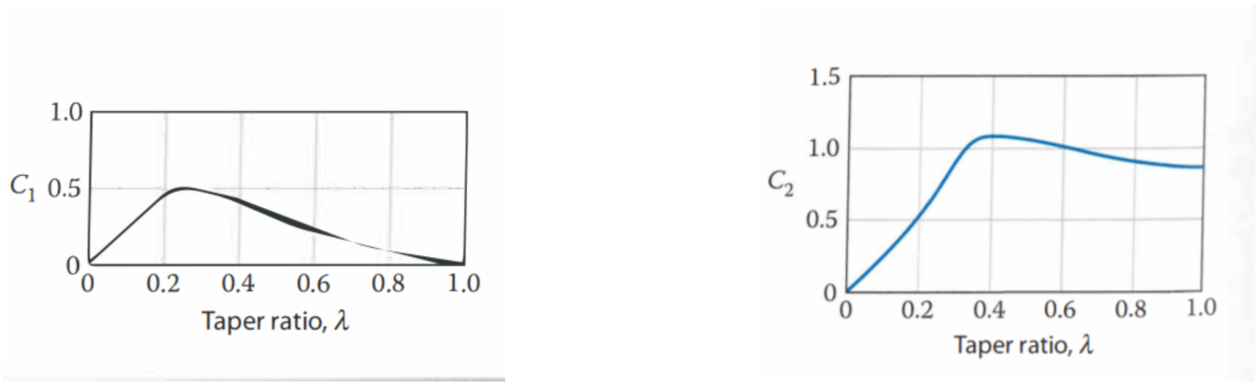



Figure 11: Taper ratio correction factors [20].

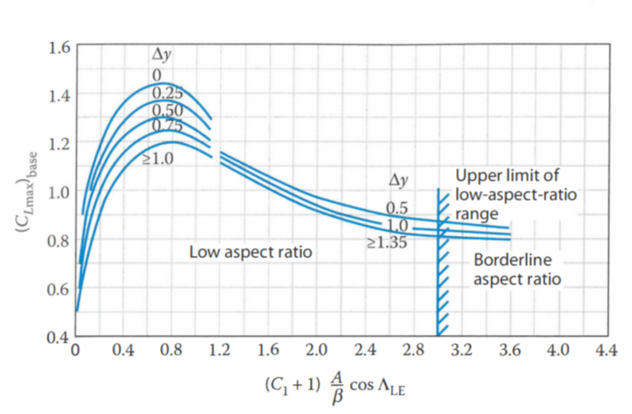



Figure 12: Maximum subsonic lift of low aspect ratio wings [20].

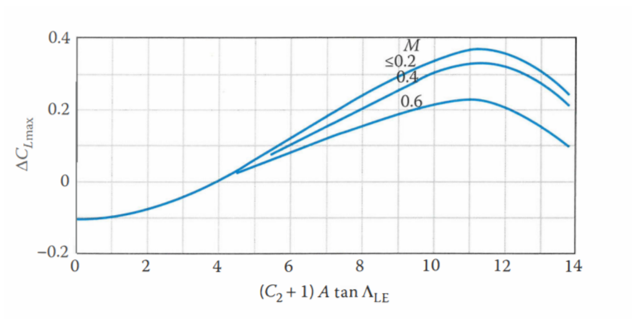



Figure 13: Maxium lift increment for low aspect ratio wings [20].

# D Heat Flux Across *Portal*

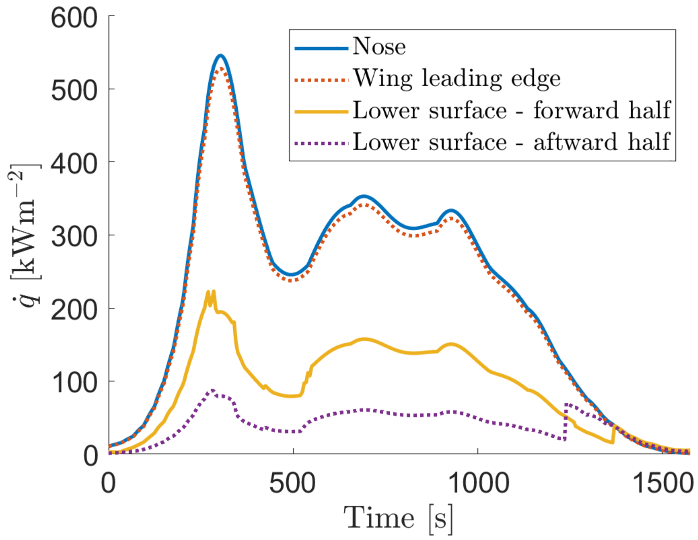



Figure 14: Maximum heat flux on surfaces of *Portal* during the optimised trajectory.